

COLOR

Color Grading Demystified

Log, LUTs & the Filmic Look — Without the Guesswork

Story Mode Guy

storymodeguy.com

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Log, LUTs & the Filmic Look — Without the Guesswork. Written for shooters whose footage looks flat, grey, or inconsistent.

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Your Footage Isn't Broken — It's Just Uncooked

You bought a camera that shoots log, watched a tutorial that promised "cinematic," dragged a LUT onto your clip, and it still looks flat, grey, or weirdly green. The problem isn't your camera and it isn't your eye — it's that nobody told you color happens in two separate stages, in a fixed order, using tools that measure the image instead of guessing at it. This guide fixes that.

Here is the whole thing stated plainly: **you correct first, then you grade.** These are two different jobs, and doing them out of order is the single most common reason footage looks amateur.



Color Correction

Making it right. Neutral exposure, accurate white balance, proper black and white points, believable skin. A clean, honest image that could go anywhere. No mood yet.



Color Grading

Making it feel. Warm and nostalgic, cold and tense, teal-shadowed and moody. Style layered *on top of* a corrected image — never in place of one.

Why the order is non-negotiable

A creative "look" is a set of adjustments calibrated to expect a normal image underneath. Apply it to raw log footage — which is flat, low-contrast, and desaturated on purpose — and the look has nothing sane to work with. The shadows are already lifted, the colors already muted, so it pushes an already-broken starting point further out of whack. That's the green, plasticky, "why does this look worse" moment everyone hits.

You cannot style an image you haven't first made true.

Correct first and the same look drops on cleanly, because now it's adjusting a normal image the way it was designed to. This is why two editors can use the identical LUT and get wildly different results — one corrected first, one didn't.

THE WHOLE GUIDE IN ONE SENTENCE

Correct the image until it's neutral and true, *then* style it — and use the scopes to tell you when each step is done, because your eyes and your monitor will both lie to you.

The pipeline you'll build

Read Chapter 2 first — the scopes are the language everything else is written in. Then work the chapters in order on a real clip:

1

Read the image with scopes.

Learn what a correct frame looks like as data, not just to your (lying) eyes.

2

Correct exposure, contrast, and white balance.

Set black and white points on the waveform; neutralize casts on the parade.

3

Convert log if needed, then build the primary grade.

Get to a normal image, then shape shadows, mids, and highlights into a base look.

4

Protect skin, isolate regions, match shots, and export right.

The finishing that makes a sequence feel like one film.



Resolve first, Premiere alongside

Every technique here is described for **DaVinci Resolve** (the free version does all of it) because its color tools are the industry standard. Where Premiere's **Lumetri** panel does the same job under a different name, we say so. Nothing depends on a specific software version or menu location — we describe the tool and what it does.

CHAPTER 2

Read the Image With Scopes

Here is an uncomfortable truth: **you cannot trust your eyes, and you almost certainly cannot trust your monitor either.** Your eyes adapt — stare at a slightly orange image for thirty seconds and your brain quietly decides it's neutral. Your laptop screen is probably too blue, too bright, and uncalibrated, so grading by eye on it makes footage look wrong on every screen but yours. Scopes are the fix: they measure the actual pixel values and never adapt, fatigue, or lie.

REMEMBER

Grade with your eyes for taste, but confirm with the scopes for truth — the scopes are the only thing in the room that isn't fooling you.

The three scopes that matter

Waveform — your exposure map

The waveform plots brightness (luminance) against the horizontal position of the frame. The bottom is pure black (0), the top pure white (1023 on a 10-bit scale, or 100 IRE). Tall traces mean bright pixels there; low traces mean dark ones. This is how you see exposure as data — what's crushing to black, what's clipping to white, and where your midtones sit.

RGB Parade — your color balance map

The parade is three waveforms side by side, one each for red, green, and blue — the single most useful scope for correction. On something that should be neutral (grey, white, or black), the three channels line up at the same height. If blue sits higher than red and green in the shadows, you have a blue shadow cast — one you literally cannot see because your eyes have adapted to it.

Vectorscope — your hue and saturation map

A circular graph: distance from center is saturation (how vivid), angle around the circle is hue (which color). A neutral grey image is a dot dead center; the more color, the more the trace spreads outward. It has one feature that will save your footage over and over — a diagonal **skin-tone line** that all human skin, of every ethnicity, falls along when correct. More in Chapter 7.



Find them fast

In Resolve, the scopes live on the Color page — open the scope panel and you can switch between Waveform, Parade, Vectorscope, and Histogram, or view all at once. In Premiere, they're the **Lumetri Scopes** panel; right-click inside it to choose which scope and which scale (use the same 0–1023 or 0–100 range consistently). Keep at least the parade visible at all times.



Set target values, not vibes

Pros don't say "that looks about right." They say "shadows sitting around 5–10, highlights topping out near 90–100, no clipping." Learn to give exposure a number. Numbers are repeatable across shots; vibes are not, which is exactly why your clips don't match.



Try this now

Open any clip and just watch the parade while you scrub. Find a shot with a white wall or a grey road and notice whether the three channels line up or one runs high. You've just spotted a color cast your eyes were hiding from you. That's the whole skill — everything else is acting on what the scopes tell you.

CHAPTER 3

Correction Step 1 — Exposure & Contrast

Correction starts with brightness, because color decisions made on a wrongly-exposed image are decisions you'll have to redo. Your first job is to set the **black point** and **white point** — anchoring the darkest and brightest parts of the frame so the image has a full, honest tonal range.

Set the black point and white point on the waveform

1 Open the waveform (or parade) and find your darkest real shadow.

Use the **Lift** wheel (Resolve) or the **Shadows/Blacks** control (Lumetri) to bring the bottom of the trace down until your true blacks sit just above 0 — around 5 to 10, not slammed to the floor.

2 Find your brightest real highlight.

Use the **Gain** wheel (Resolve) or **Highlights/Whites** (Lumetri) to lift the top of the trace up toward 90–100 without clipping — unless it's a light source or specular sparkle, which is allowed to blow out.

3 Adjust the midtones last.

Use the **Gamma** wheel (Resolve) or **Exposure/Midtones** (Lumetri) to place the overall brightness where the image reads correctly. Faces should generally sit in the upper-middle of the waveform.

Healthy exposure targets

STARTING POINT

Black point **5-10**

Skin/midtones **~50-70**

White point **90-100**

Clipping **only light sources**

Values on a 0–100 IRE scale. These are targets to correct *toward*, then bend for creative effect afterward.



Don't crush your blacks

Slamming the shadows all the way to 0 to make an image "punchy" throws away detail you can never get back — texture in dark clothing, shadow gradients, the roll-off that makes footage feel filmic. Crushed blacks are the number-one giveaway of an over-eager beginner. Leave a little life at the bottom; you can always take it down later for a specific look.

Build contrast — but as a shape, not a slider

Contrast is just the distance between your black point and white point plus how the midtones curve between them. Once your points are set, a gentle **S-curve** on the custom curve tool is the most filmic way to add contrast: pull the shadows down a touch and lift the highlights a touch, leaving the middle anchored. This deepens the image without the harsh, video-y look of just cranking a contrast slider.



Highlight roll-off is the "film" feeling

Digital tends to hit its white ceiling abruptly, which reads as harsh. A soft S-curve that eases the brightest tones as they approach the top — rather than a hard clip — is a big part of what people mean when they say footage looks "cinematic" versus "like video." Roll the highlights, don't wall them.

Correction Step 2 — White Balance

With brightness honest, fix color. White balance means making things that should be neutral — a white shirt, a grey wall, a cloudy sky — actually read as neutral instead of tinted orange, blue, or green. The RGB parade is your instrument, because it shows a cast as a plain height difference between channels that no amount of eye-adaptation can hide.

Neutralize a cast with the parade

1 Find something in frame that should be neutral.

A white wall, grey pavement, the whites of an eye, a silver object. This is your reference.

2 Watch that area on the parade.

If red runs high and blue runs low, the shot is too warm. If blue runs high, too cool. If green runs high, that's the ugly fluorescent/log-green cast.

3 Balance the channels at each range.

Nudge the **Lift** wheel to align the three channels in the shadows, **Gamma** to align them in the mids, and **Gain** to align them in the highlights. When all three traces sit level on neutral objects, your cast is gone.



Let the eyedropper do the first pass

Resolve and Lumetri both have a white-balance **eyedropper/picker**. Click it on something that should be neutral grey and the software auto-balances the channels for you. Use it as a fast starting point, then refine on the parade by eye — the auto-pick gets you 80% there in one click.



Balancing the shadows and highlights separately

A shot can be warm in the highlights but cool in the shadows — common under mixed lighting, like tungsten lamps plus window daylight. That's why you balance Lift, Gamma, and Gain independently instead of using one global temperature slider. Aligning all three ranges is what makes a mixed-light shot finally look clean.



Neutral is the target, not the destination

You are correcting to neutral so you have a true baseline — you are *not* committing to a neutral final look. A cozy interior *should* feel a little warm in the end. But add that warmth deliberately in the grade, on purpose, on top of a neutral base — don't leave in an accidental cast and call it a style.

CHAPTER 5

Log Footage & LUTs — What's Actually Going On

If your footage looks washed-out, grey, and lifeless straight off the camera, and you shot in a mode called Log (S-Log, C-Log, V-Log, Log3, HLG's cousin), that flatness is **on purpose** and it's a good thing. Understanding why is what finally makes LUTs make sense.

What log is and why it looks flat

A camera sensor captures a wide range of brightness — deep shadows to bright highlights — but a normal video file can only store so much before it clips. Log recording squeezes that huge range into the file with a logarithmic curve: it lifts shadows, pulls down highlights, and mutes colors so nothing gets thrown away. It looks flat and grey because the contrast and saturation have been *deliberately compressed and stored for later*. Log is a raw ingredient — maximum information, no seasoning.

Log footage isn't ugly. It's undeveloped — all the range is there, waiting for you to expand it back out.

Conversion (technical) LUTs vs look (creative) LUTs

A LUT — Look-Up Table — is just a recipe that maps every input color to an output color. There are two fundamentally different kinds, and confusing them is the root of most LUT frustration.

	Conversion / Technical LUT	Look / Creative LUT
Job	Undo log — expand it back to a normal, neutral image	Add a creative mood on top of a normal image
Example names	"S-Log3 to Rec.709", "LogC to Rec709"	"Teal Sunset", "Vintage Film 03"
When to apply	First, to get to a correct baseline	Last, after correction
Interchangeable?	No — must match your exact camera/log profile	No — expects a normal image underneath

A **conversion LUT** is the "develop" step that turns your specific log profile back into a standard Rec.709 image. It must match your camera and log format exactly — an S-Log3 clip needs an S-Log3-to-Rec709 LUT, not a Canon one. Apply it early and grey footage snaps back to a normal starting point. A **look LUT** is creative — someone's pre-baked mood, designed to sit on a *normal* Rec.709 image, not on raw log.



Why a look LUT alone fails

Drop a creative "look" LUT straight onto uncorrected log and it has to fight the flat, lifted, desaturated starting point it was never designed for. You get muddy shadows, weird green skin, and blown, oversaturated highlights. The look isn't broken — it's being asked to style an image that was never developed. Convert (or correct) first, then the same look drops on beautifully.

The correct order with log

1

Apply the conversion LUT (or convert manually).

Use your camera's matching log-to-Rec709 LUT, or use Resolve's Color Space Transform, to get to a normal image. In Resolve this is often better handled by setting the project's **Color Management** to interpret your log input automatically.

2

Do your correction on top.

Fine-tune exposure and white balance as in Chapters 3 and 4 — the conversion gets you close, but every shot still needs balancing.

3

Now grade, or apply a look LUT.

With a corrected image underneath, creative looks finally behave.



Dial back any LUT with one slider

Both Resolve and Lumetri let you reduce a LUT's **intensity/mix**. Look LUTs are almost always too strong at 100%. Apply it, then pull it back to 40–70% until it enhances the image instead of steamrolling it. A LUT is a starting point, never the finished grade.

CHAPTER 6

The Primary Grade — Lift, Gamma, Gain

Correction is done: the image is neutral and true. Now comes the fun part — the **primary grade**, where you shape the whole frame into a base look using the three color wheels. Master these and you can build almost any mood.

What each wheel actually controls

● Lift = Shadows

Moves your darkest tones. Push it blue for cool, moody shadows; nudge it warm for a softer black. This is where a lot of "mood" lives.

◐ Gamma = Midtones

Moves the middle — where most of your subject and skin sit. Small moves here have the biggest visible effect on overall feel.

○ Gain = Highlights

Moves your brightest tones. Warm gain gives golden, sunny highlights; cool gain gives a clean, clinical light.

In Resolve these are the three color wheels on the Color page (with a fourth, **Offset**, that shifts everything at once). In Premiere's Lumetri, the same idea lives in the **Color Wheels** section (Shadows / Midtones / Highlights) plus the temperature and tint sliders.

A reliable base-look recipe

1 Set your contrast and pivot first.

Confirm the S-curve from Chapter 3 is giving you clean depth before adding color.

2 Choose a shadow direction.

Most cinematic looks push shadows slightly cool. A tiny blue-teal nudge in Lift adds instant atmosphere. Keep it subtle.

3 Warm the highlights or midtones slightly.

A touch of warmth where the light and skin live makes people look healthy and the scene inviting.

4 Set overall saturation.

Bring saturation to taste — usually a modest lift. Watch the vectorscope so you don't oversaturate into cartoon territory.

Warm, natural base look

STARTING POINT

Lift **slightly cool**

Gamma **neutral / faint warm**

Gain **slightly warm**

Saturation **+5 to +15**

A safe, flattering starting grade for people and daylight scenes. Small moves — you should barely see the wheels leave center.



The orange-and-teal cliché

Pushing highlights hard orange and shadows hard teal is the most overused grade of the last fifteen years. A little of that complementary split is genuinely flattering; a lot of it screams "I just discovered color wheels." If your footage looks like a blockbuster movie poster, dial it back by half. Restraint reads as taste.



Grade in nodes, not one pile

In Resolve, keep separate **nodes** for each job: one for the log conversion, one for exposure/balance correction, one for the primary look, one for skin, one for windows/masks. If a shot looks off later, you can fix the exact stage responsible instead of untangling everything. In Premiere, stack multiple Lumetri instances for the same reason — one effect per job.

CHAPTER 7

Skin Tones — Keep People Looking Human

Audiences forgive almost any stylistic choice except one: sickly-looking skin. Green, orange, purple, or grey faces break the shot instantly, because everyone knows what people are supposed to look like. Fortunately, healthy skin has a precise, measurable target — and it's the same target for every human being on Earth.

The skin-tone line on the vectorscope

The vectorscope has a diagonal reference line running from the center toward the upper-left, sometimes labeled "I" or "skin tone." Here's the remarkable part: **correctly-lit skin of every ethnicity falls along this same line.** Darker and lighter skin differ in *saturation* — how far the trace extends along the line — but the *angle* is the same for everyone. It's a property of blood under skin, and it's astonishingly consistent.

REMEMBER

If your subject's skin trace is drifting off the skin-tone line on the vectorscope, they will look unwell — pull it back onto the line and they look human again, every time.

How to check and fix skin

1

Isolate the face if you can.

Use a temporary window/mask or a qualifier (Chapter 8) so the vectorscope shows mostly skin, not the whole busy frame.

2

Look at the angle of the skin trace.

It should point along the skin-tone line. Too far clockwise reads green/yellow; too far counter-clockwise reads pink/magenta.

3

Nudge hue and balance until it sits on the line.

Small pushes to your midtone balance or a hue adjustment bring skin back onto target. Fix the whole-image cast first (Chapter 4) — often skin snaps onto the line the moment your overall white balance is right.



Green skin is the log tell

If faces look faintly green, it's usually leftover log/fluorescent cast that correction missed. Take a little green out of the midtones (add magenta) and watch the skin trace rotate back onto the line. This one move rescues more footage than any creative grade.



Don't over-warm to "fix" skin

Beginners drown a face in orange to make it look "healthy" and end up with an Oompa-Loompa. The target is the vectorscope line, not "more orange." Trust the scope over the impulse to keep cranking warmth.

CHAPTER 8

Secondary & Selective Color

Everything so far was *primary* grading — it affects the whole frame at once. **Secondary** grading affects only part of the frame: just the sky, the skin, the red jacket. This is how you make one element pop, tame a distracting color, or fix skin without touching the background. Two tools do the isolating.

Qualifiers (keys) — isolate by color

A **qualifier** (Resolve) or **HSL Secondary** (Lumetri) lets you pick a color with an eyedropper and grade only pixels that match it. Click the sky and you can deepen just the blue; click skin and you can smooth just the faces. You then refine the selection with hue/saturation/luminance sliders until only the thing you want is selected.

1 Pick the color with the eyedropper.

Click the element you want to isolate — sky, skin, a green wall.

2 Turn on the selection highlight.

View the matte in black-and-white so you can see exactly what's selected. Widen or narrow the hue/sat/luma ranges until only your target is white.

3 Blur/soften the edge slightly.

A little softening on the key prevents a hard, cut-out look. Then grade — shift the hue, drop or boost saturation, brighten or darken just that region.

Power windows & masks — isolate by shape

A **power window** (Resolve) or **mask** (Lumetri) isolates a region by drawing a shape — a circle, a rectangle, a free-drawn curve — rather than by color. Use it to darken the corners of the frame (a subtle vignette that pulls the eye to your subject), brighten a face, or hold back an overexposed window.



Track the window so it follows motion

If your subject moves, a static window will slide off them. Both Resolve and Lumetri have **tracking** — analyze the motion and the window follows the subject through the shot automatically. This is how you keep a face-brightening mask locked on someone as they walk across frame.



Combine a window and a qualifier

Want to boost only the sky, but the eyedropper also grabs a blue shirt? Draw a power window around the upper part of the frame and put the qualifier inside it. Now you're selecting "blue pixels, but only up here." Shape plus color together is how you isolate almost anything cleanly.



Subtlety is the whole game

Over-cooked secondaries — a radioactive blue sky, a vignette you can see the edge of, skin smoothed into plastic — look worse than doing nothing. The best secondary work is invisible; the viewer feels the shot is clean and their eye lands where you want, without knowing why.

CHAPTER 9

Matching Shots Across a Scene

A single graded shot can look gorgeous and still ruin your edit if the next shot looks nothing like it. Nothing screams "amateur" louder than a conversation where one angle is warm and bright and the reverse angle is cool and dark. **Shot matching** is the workflow that makes a whole sequence feel like one continuous world.

The hero-shot workflow

1

Grade your best "hero" shot first.

Pick the most important or best-lit shot in the scene, correct and grade it fully, and make it your reference for everything else.

2

Match every other shot to the hero using scopes.

Put the hero and the next shot side by side. Compare their parades: line up the black points, the white points, and the channel balance so both read the same. Match the numbers, not just the feel.

3

Correct each shot to neutral, then copy the look.

Balance each new shot on its own first, *then* apply the same creative grade. This is why correction-before-grading matters at sequence scale: a shared look only lands consistently on shots that were each neutralized first.



Use the split-screen and stills

Resolve lets you **grab a still** of your hero shot and wipe it against the current clip for a direct side-by-side comparison. Premiere users can use a comparison view or simply toggle between clips. Matching against a reference image beats matching from memory every time — your eyes forget the last shot within seconds.



Match faces and neutrals, not the whole frame

Two shots of the same scene may frame different backgrounds, so don't try to make the whole image identical. Match the things that *should* be constant: skin tone on the same person, a white shirt, the color of the light. Get those consistent and the eye reads the whole scene as matched, even if the backgrounds differ.



Beware the auto-match button

Resolve's **Shot Match** and similar auto tools can get you a rough starting point, but they frequently overshoot or misread mixed lighting. Treat any automatic match as a first draft to refine on the scopes — never as the final answer.

CHAPTER 10

Save the Look & Export Right

You've corrected, graded, protected skin, isolated regions, and matched your scene. Two things remain: **saving your look so you can reuse it**, and **exporting so it survives the trip to YouTube, Instagram, and everyone's phone** without turning washed-out or dark.

Save and reuse your grade

Export a still / grade

In Resolve, save a graded shot as a **still** (in the gallery) or copy the grade from clip to clip. This carries your exact node tree to the next shot instantly.

Export a LUT

Once you've built a look you love, export it as a **.cube LUT**. Now you can reuse that exact grade on future projects — or drop it in Premiere. Your own signature look, bottled.

Export the look on a clean node

When you export a LUT, remember it captures the color transform, not shot-specific corrections. Build your reusable look on a node that assumes an already-corrected image — so the LUT stays a "look" and doesn't bake in one clip's exposure fix.

Deliver in Rec.709 so it doesn't wash out

Here is the last trap, and it catches almost everyone: you finish a beautiful grade, export, upload — and online it looks flat, grey, or washed-out. The cause is a **color space mismatch**. Standard web video expects **Rec.709**, the reference standard for normal (non-HDR) video. If your project or export isn't set up for Rec.709, players and phones misinterpret your colors and the contrast falls apart.

REMEMBER

Grade and monitor in Rec.709, and export in Rec.709 — matching your grading environment to your delivery target is what keeps the image you built from washing out the moment it leaves your machine.

- ✓ Set your project timeline/output color space to **Rec.709** for normal web and social delivery.
- ✓ Grade on a monitor showing Rec.709, not raw log, so what you see is what you export.
- ✓ Export with a widely-compatible codec (H.264/H.265) at a healthy bitrate — low bitrate crushes your careful gradients into banding.
- ✓ Keep the tag/flag correct: a Rec.709 file marked as something else is exactly what makes footage look faded online.
- ✓ Test on a phone before you publish — most of your audience watches there, and it's where mismatches show worst.



Don't accidentally deliver raw log

Exporting the flat log image because you forgot the conversion is the most washed-out result possible. If your export looks grey and lifeless, check that your log-to-Rec709 conversion is active on the render, not just in your viewer preview.



Your practice project this week

Take one 30-second clip and run the whole pipeline in order: read the scopes, set black and white points, balance on the parade, convert the log, build a primary look, check skin on the vectorscope, then export in Rec.709. Do it once, start to finish, and the sequence will finally be muscle memory instead of a pile of disconnected tutorials.



The habit that separates pros

Every professional colorist does the same unglamorous thing: they correct to neutral and confirm it on the scopes *before* they have any fun with the look. Build that discipline and consistency stops being luck — your footage matches, your skin looks human, and your grade survives the upload.

REMEMBER

Color isn't a filter you apply — it's a craft you build, one honest correction at a time, until the mood you felt on set is the mood on the screen.

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